Per-site ΔRMSE from adding disturbance — M3 (C+T) vs M4 (C+T+D)

Below 0 = D reduces the site's RMSE | bars coloured by relative mortality (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (106 sites, 388 site-years) · deadtrees v4.1

XGBoost (Optuna) · 12m | **XGBoost (Optuna) · 24m** | **LightGBM (Optuna) · 12m** | **LightGBM (Optuna) · 24m**

GPPsat (µmol m⁻² s⁻¹)

57% improved | 55% improved | 55% improved | 58% improved

NEPmax (µmol m⁻² s⁻¹)

59% improved | 65% improved | 68% improved | 66% improved

ETmax (mm d⁻¹)

64% improved | 58% improved | 62% improved | 58% improved

WUE (g C mm⁻¹)

58% improved | 55% improved | 52% improved | 58% improved

ΔRMSE = RMSE(with D) − RMSE(without D)

Sites (sorted by ΔRMSE within each panel)

Relative mortality (%)
site maximum, 500 m
breaks 9.5 | 19.9

50
40
30
20
10
0